

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (Electrical Engineering)

May 2012

EE 308 High Voltage Engineering

Date: 09.05.2012, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) State and explain Paschen's law. Derive expressions for $(pd)_{min}$ and $(V_b)_{min}$. [07]
- Q - 2 (a) Discuss various mechanisms of vacuum breakdowns. [08]
- (b) In an experiment in a certain gas it was found that the steady state current is 5.5×10^{-8} A at 10 kV at a distance of 0.4 cm between the plane electrodes. Keeping the field constant and reducing the distance to 0.2 cm results in a current of 5.5×10^{-10} A. Calculate Townsend's primary ionization coefficient. [03]
- (c) Explain why the breakdown strength of electronegative gas is more than air. [03]

OR

- Q - 2 (a) List out the different mechanisms by which breakdown occurs in solid dielectrics and explain any two in detail. [08]
- (b) Explain breakdown mechanism in commercial liquids. [06]
- Q - 3 Explain any two: [14]
- [1] Corona discharges
 - [2] Steamer Theory of breakdown in air at atmospheric pressure
 - [3] Oil purification system

SECTION - II

- Q - 4 (a) Explain the method of cascade connection of transformer for producing very high voltage. [07]
- Q - 5 (a) Explain the Marx circuit arrangement for multistage impulse generator. [06]
- (b) A 10-stage impulse generator has $0.12 \mu\text{F}$ condensers. The wave front and the wave tail resistances connected are 800Ω and 1000Ω respectively. If the load condenser is 1000 pF , find the front and tail times of the impulse wave produced. [03]
- (c) Explain the construction and working of electrostatic voltmeter. [05]

[P.T.O.]

OR

- Q - 5 (a) Explain the construction and working of CVT [06]
(b) List out the various methods used for measurement of high voltages and high currents. [03]
(c) Explain high voltage Schering bridge for measurement of capacitance and $\tan \delta$. [05]
- Q - 6 Attempt any two: [14]
[1] Layout of high voltage laboratory
[2] Van de Graff generator
[3] Synthetic testing of circuit breakers
